Set 6　4.3　4.13　4.16　4.17　4.18　4.19　4.24

4.3 解: (1) $V_x = \frac{V_x' + M}{1 + M\frac{V_x'}{c^2}}$

$V_x' = 0.8c$, $M = 0.5c$ $\Rightarrow V_x = 0.93c$

(2) $V_x = \frac{V_x' + M}{1 + \frac{MV_x'}{c^2}}$

$V_x' = -c$, $M = 0.5c$ $\Rightarrow V_x = -c$

4.13 解 $t_2' - t_1' = \gamma(t_2 - t_1)$ $\Rightarrow \gamma = 2$, $M = \frac{\sqrt{3}}{2}c$

$x_2' - x_1' = -M\gamma(t_2 - t_1)$ $\Rightarrow x_2' - x_1' = -3\sqrt{3} \times 10^8 m$

若 $t_2' - t_1' = 0.9s$ 则 $\gamma = \frac{9}{10} < 1$, $M > c$

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4.16 解: $x_2' - x_1' = \gamma(x_2 - x_1)$ $\Rightarrow \gamma = 2$, $M = \frac{\sqrt{3}}{2}c$

$t_2' - t_1' = -\gamma M \frac{x_2 - x_1}{c^2}$ $\Rightarrow t_2' - t_1' = -\frac{\sqrt{3}}{3} \times 10^{-8} s$

4.17 解 $\gamma = \frac{1}{\sqrt{1 - \frac{M^2}{c^2}}}$

$t_2 - t_1 = \gamma\left[(t_2' - t_1') + M\frac{(x_2 - x_1')}{c^2}\right]$ $\Rightarrow t_2 - t_1 = 4 \times 10^{-14} s$

车尾先击中

4.18 解: $V_x' = \frac{x_2' - x_1'}{t_2' - t_1'} =$ $\Rightarrow V_x' = 2.5 \times 10^8 m/s$

$\gamma = \sqrt{1 - \frac{M^2}{c^2}}$ $\gamma = \frac{5}{4}$

$x_2 - x_1 = \gamma\left[(x_2' - x_1') + M(t_2' - t_1')\right]$ $\Rightarrow x_2 - x_1 = 215m$

$t_2 - t_1 = \gamma\left[(t_2' - t_1') + M\frac{(x_2' - x_1')}{c^2}\right]$ $\Rightarrow t_2 - t_1 = 7.5 \times 10^{-7} s$

$V_x = \frac{x_2 - x_1}{t_2 - t_1}$ $\Rightarrow V_x = 2.87 \times 10^8 m/s$

4.19 解: (1) $V_x' = \frac{V_x - M}{1 - \frac{MV_x}{c^2}}$ $\Rightarrow V_x' = -0.946c$

$V_x' = -2.84 \times 10^8 m/s$

(2) $\Delta t' = \frac{\Delta t}{\gamma}$ $\Delta t = 4s$

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4.21

(1) $\begin{cases} l = \dfrac{l_0}{\gamma} \\ m = m_0\gamma \\ \lambda_1 = \dfrac{m}{l} = \cdots \end{cases} \Rightarrow \lambda_1 = \dfrac{m_0}{l_0}\left(1 - \dfrac{u^2}{c^2}\right)$

(2) $\begin{cases} l = l_0 \\ m = m_0\gamma \\ \lambda_2 = \dfrac{m}{l} \end{cases} \Rightarrow \lambda_2 = \dfrac{m_0}{l_0}\sqrt{1 - \dfrac{u^2}{c^2}}$

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